

Adaptive Staircase Procedure

PSY310: Lab in Psychology

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GitHub Link

<https://github.com/HetanshJalan/Psychology-Experiments>

Introduction

The Adaptive staircase procedure is a simple method used in psychophysics to determine the participant's perceptual threshold. The procedure does not include a series of stimulus intensities, but rather the stimulus is adjusted depending on the previous response of the participant. If correct responses are made, the staircase goes up, if incorrect responses are made, the staircase goes down. In the present task the adaptive parameter was the amplitude of a tilted sinusoidal grating as compared to vertical. Participants responded to a direction of tilt of the grating as a two-alternative forced choice (2AFC) response (left or right tilt). The first part of the staircase had a large tilt and discrimination was easy, followed by a decrease in tilt for correct responses. Next, it was desired to know at what angle the orientation discrimination became unreliable when the angle was large. It's the changed up-down logic that is typical of a psychophysical approach (as outlined by Levitt 1971), and is usually applied to converge to 80% correct performance (PsychoPy 2026a, 2026b).

Aim- An adaptive 3-down/1-up staircase procedure to determine a participant's orientation-discrimination threshold.

Hypothesis- The more the tilt decreases, the less accurate the discrimination will be. The stairs should thus be made to approach a narrow band of tilt magnitudes around the participant's discrimination threshold.

Method

Participant

The experiment was carried out on a single 20-year-old male undergraduate student. Student researcher was the participant. There were no mandatory requirements for participation and data were collected under the pilot label.

Materials

The experiment was created and run in PsychoPy 2026.2.3. System information: laptop is running a processor of 11th generation Intel Core i3.

The system includes i5-1135G7 processor (2.40 GHz), memory 16 GB RAM, NVIDIA graphics card: GeForce MX350 (2 GB) and Windows 11 Home Single Language, version 25H2. The

average frame rate of the display during the run was about 59.85 frames per second (fps). The answers were collected with the help of the computer keyboard.

Gaussian masked sinusoidal grating was used as the stimulus. The size of the grating was (0.3, 0.3) PsychoPy height and the spatial frequency of the grating was 8 cycles per stimulus unit. It was either +level or -level (where level is the size of the tilt to adapt to, in degrees). The fixation cross was a white cross (500ms) shown before each grating. The grating was presented for a given period of time (300ms). The answer part provided for answers was left and right.



Figure 1. Fixation cross presented for 500 ms before the orientation stimulus.

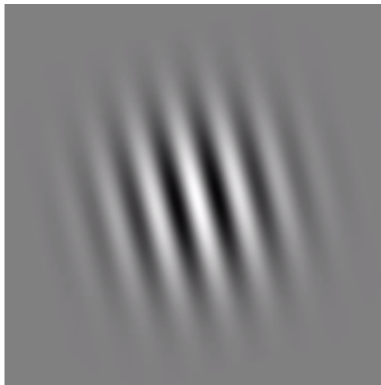


Figure 2. Left-oriented tilted grating stimulus.

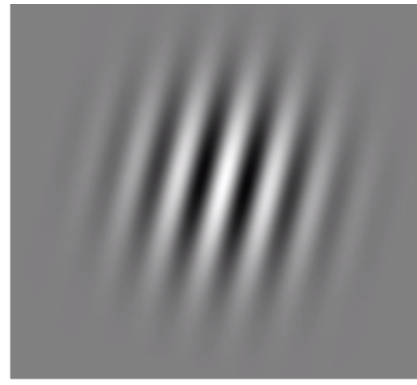


Figure 3. Right-oriented tilted grating stimulus.

Staircase parameters

Parameter	Setting
Start value	10° tilt magnitude
Minimum / maximum	1° / 20°
Step type	Linear
Step sizes	2°, 1°, 1°, 0.5°
N up	1 incorrect response
N down	3 correct responses
Minimum trials	50

Response mapping	Left = -tilt; Right = +tilt
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Procedure

The participant was seated at the computer and asked to evaluate as rapidly and accurately as possible the direction of the tilt of the grating. All trials started with a white fixation cross in the center for 500 msec. Then a sinusoidal grating (300 ms) was presented at an orientation corresponding to the level of the staircase at the time of its onset. The sign of the orientation was counterbalanced across trials: on half the trials the negative orientation was associated with a left response, and on the other half of the trials, the positive orientation was associated with a right response. The keyboard response interrupted the trial and it was noted correct or incorrect. Each answer elicited was followed by an adjustment of the next tilt magnitude by the staircase following the programmed rule. The starting angle for the staircase was 10° and the range was between 1° and 20°. It was used for 50 trials.

Data treatment

For the 50 completed trials, the CSV output was verified for response accuracy, tilt magnitude, step size and reaction time. The accuracy was determined as the percentage of correct responses made across the trials. All reaction times were converted to milliseconds for reporting purposes. Reversal points were determined based on the changes in the direction of the staircase trajectory. The reversal intensities were summarised descriptively and an inferential test was not run due to the small sample size of one participant.

Results

50 trials were obtained from the staircase. The accuracy of the participant was 46 of 47, or 92.0%. Mean reaction time was 621.9 ms (SD = 695.4 ms), with a median of 452.6 ms.

Staircase convergence

The stimulus was presented at a 10° and was raised by 2° after the first error. The staircase then descended as progression of correct responses was made. Seven reversals were observed at tilt magnitudes of 11°, 8°, 7.5°, 7.5°, 7°, 7.5°, 7°. The average value of all seven reversal intensities is 7.93°. The later reversals (reversals 3-7) followed the larger adjustment and concentrated closely around 7.3° with a mean value of 7.30°. The average of the last 4 reversals was 7.25°. These values suggest that the staircase was tested later in the range 7–7.5°.

Table 1

Observed performance at each tested tilt magnitude

Tilt (°)	Trials	Correct	Accuracy	Mean RT (ms)
4.5	1	1	100 %	512.0
5.0	3	3	100 %	554.2
5.5	3	3	100 %	453.9
6.0	3	3	100 %	479.2
6.5	3	3	100 %	485.4
7.0	11	8	72.7 %	451.5
7.5	9	9	100 %	682.1
8.0	6	6	100 %	408.7
9.0	3	3	100 %	419.2
10.0	4	3	75.0 %	1532.6
11.0	3	3	100 %	569.1

12.0	1	1	100 %	2010.1
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The 7.0° level was presented 11 times and produced 8 correct responses (72.7%), whereas 7.5° was presented 9 times and all responses were correct. These observations are descriptive because the staircase was designed to adapt its sampling around the participant's performance.

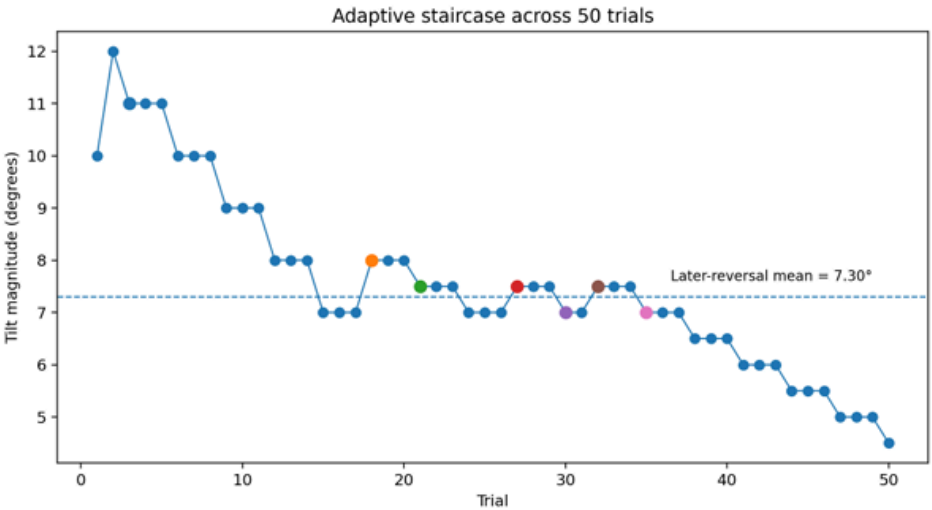


Figure 4. Trial-by-trial change in tilt magnitude. Markers indicate reversal points; the dashed line marks the mean of the later five reversals (7.30°).

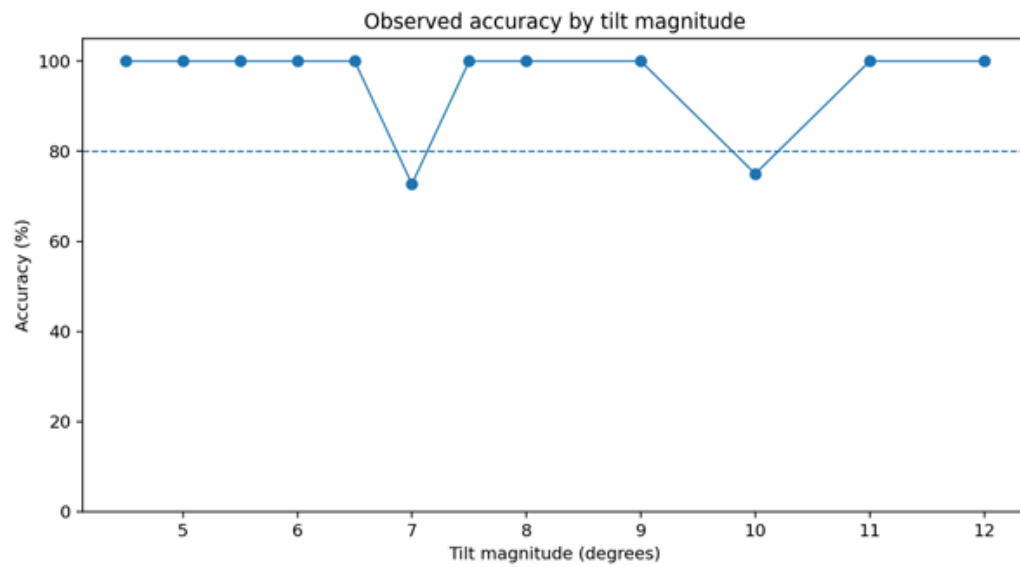


Figure 5. Observed accuracy by tilt magnitude. The dashed line marks 80%, the approximate convergence level targeted by a 3-down/1-up staircase.

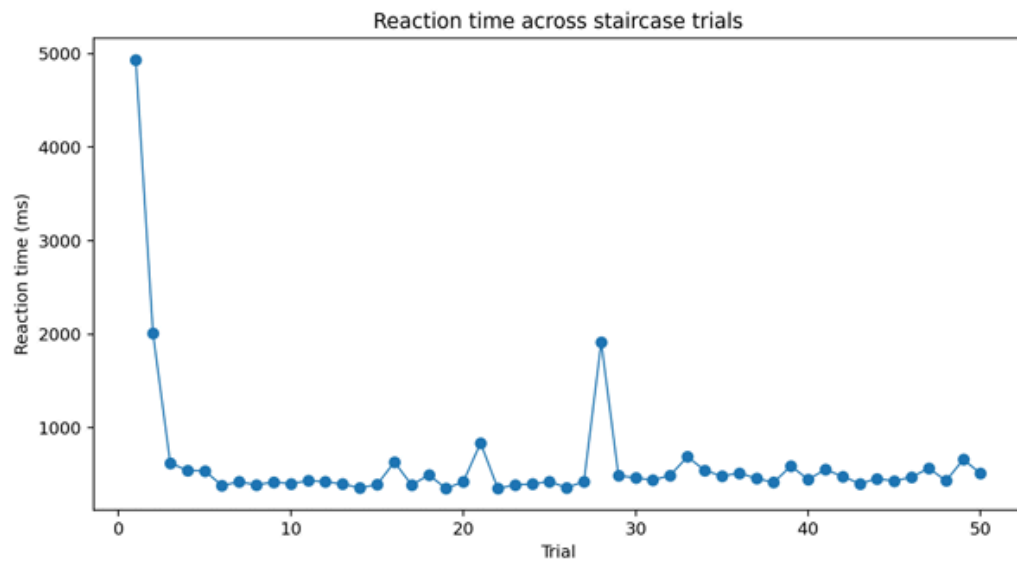


Figure 6. Reaction time across the 50 staircase trials.

Reversal points

Reversal	Trial	Tilt (°)	New direction
1	3	11.0	decreasing
2	18	8.0	increasing
3	21	7.5	decreasing
4	27	7.5	increasing
5	30	7.0	decreasing
6	32	7.5	increasing
7	35	7.0	decreasing

Discussion

The adaptive pattern of the staircase was as expected: large tilts were generally discriminated correctly, and the more often a correct response was given, the smaller the tilt. Occasional incorrect responses then led to an increase again when the stimulus was at the more challenging level for the participant. The later reversals tended to be concentrated between 7° and 7.5° indicating that the participant's orientation discrimination had been tested most closely there.

The programmed 3 down/1 up rule is designed to strive for an 80% accurate level. The subject was correct in 72.7% of the trials presented at 7.0° and 100% of the trials presented at 7.5° in the present data. Later-reversal mean of 7.30° is thus a useful descriptive estimate of the threshold region, not a population level threshold. The outcome is in keeping with the design of an adaptive staircase: It focused observations on a difficulty level rather than randomizing all 50 trials between a 1° and 20° range.

Reaction time was not a primary outcome, but it does give a secondary description of performance. The average time of reaction was 621.9ms, and the standard deviation was quite large, 511.9ms.

695.4 ms. This variability was partially due to a few long responses, such as the first trial (around 4.94 s) and the second trial (around 2.01 s). An unacceptably large amount of reaction-time variability should not be considered a threshold indicator, as the staircase was designed for accuracy, not speed.

The biggest drawback is the single participant design. This later-reversal value of 7.30° is a best estimate for this participant's condition with this stimulus and hardware and cannot be extrapolated to a larger population. There were seven reversals over the 50 trials, but the last trials were not completed at a pre-specified stable threshold estimate, as in the staircase procedure. Multiple participants would have been a better design, with a fixed number of stable reversals, a fixed number of final reversals, and a preset rule for calculating the final threshold from the final reversals. Repeated runs could also be used to assess within-participant reliability.

Conclusion

The adaptive staircase was able to successfully alter the tilt of the grating in response to the participant's responses over 50 trials. The later reversal points were closer together, and the mean of the later reversals was 7.30° (the later-reversal mean was around 7.0° – 7.5°). The data thus give a descriptive estimate of the status of the participant,

The single-participant design and limited number of reversals limit the strength and generalizability of the conclusion, and the orientation-discrimination threshold region is not discussed.

Appendix A: Trial-level data

The table below reproduces the main trial-level variables from the recorded CSV output. Reaction time is shown in milliseconds; 1 = correct and 0 = incorrect.

Trial	Tilt (°)	Step	Response	Correct	RT (ms)
1	10.0	2.0	Right	0	4935.3
2	12.0	1.0	Right	1	2010.1
3	11.0	1.0	Right	1	623.0
4	11.0	1.0	Right	1	544.6
5	11.0	1.0	Right	1	539.9
6	10.0	1.0	Right	1	382.7
7	10.0	1.0	Right	1	423.1
8	10.0	1.0	Right	1	389.3
9	9.0	1.0	Right	1	421.4
10	9.0	1.0	Right	1	401.7
11	9.0	1.0	Right	1	434.4
12	8.0	1.0	Right	1	423.9

13	8.0	1.0	Right	1	398.4
14	8.0	1.0	Right	1	359.3
15	7.0	1.0	Right	1	392.6
16	7.0	1.0	Right	1	634.4
17	7.0	1.0	Right	0	389.7
18	8.0	1.0	Right	1	495.1
19	8.0	1.0	Right	1	351.7
20	8.0	0.5	Right	1	423.6
21	7.5	0.5	Right	1	838.2
22	7.5	0.5	Right	1	350.8
23	7.5	0.5	Right	1	391.2
24	7.0	0.5	Right	1	399.0
25	7.0	0.5	Right	1	425.9
26	7.0	0.5	Right	0	359.7
27	7.5	0.5	Right	1	426.7
28	7.5	0.5	Right	1	1915.6
29	7.5	0.5	Right	1	488.0
30	7.0	0.5	Right	1	466.1
31	7.0	0.5	Right	0	439.7
32	7.5	0.5	Right	1	492.6
33	7.5	0.5	Right	1	690.2
34	7.5	0.5	Right	1	545.4
35	7.0	0.5	Right	1	483.9
36	7.0	0.5	Right	1	516.4
37	7.0	0.5	Right	1	459.1
38	6.5	0.5	Right	1	416.2
39	6.5	0.5	Right	1	591.7

Trial	Tilt (°)	Step	Response	Correct	RT (ms)
40	6.5	0.5	Right	1	448.2
41	6.0	0.5	Right	1	554.0
42	6.0	0.5	Right	1	480.8
43	6.0	0.5	Right	1	403.0
44	5.5	0.5	Right	1	457.1
45	5.5	0.5	Right	1	430.3
46	5.5	0.5	Right	1	474.5
47	5.0	0.5	Right	1	567.6

48	5.0	0.5	Right	1	434.1
49	5.0	0.5	Right	1	661.0
50	4.5	0.5	Right	1	512.0

References

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